Lab 5: Examining the Registry

What You Need for this lab

- Install Virtualbox : https://www.virtualbox.org/wiki/Downloads
- Install Kali 2021.4. : https://old.kali.org/kali-images/kali-2021.4/
  - Notes: Suggest You configure the disk size of Kali VM 80G because the size of each leakage cases image is 30G+
  - Add new virtual hardisk size 20 GB
- Tools RegRipper 3.0
- Download DD images
  - https://www.dropbox.com/s/u4axlx8eomgwfdc/cfreds_2015_data_leakage_pc.7z.001
  - https://www.dropbox.com/s/pyq50s2cri6yftf/cfreds_2015_data_leakage_pc.7z.002
  - https://www.dropbox.com/s/cxvzuiupmqc7l99/cfreds_2015_data_leakage_pc.7z.003

Step 1.

Unzipped the DD image

```
root@kali:~/lab# 7z e cfreds_2015_data_leakage_pc.7z.001
```

Verify the unzipped DD image

```
root@kali:~/lab# ls -l
total 26272120
-rw-r--r-- 1 root root  2147483648 Apr 29  2015 cfreds_2015_data_leakage_pc.7z.001
-rw-r--r-- 1 root root  2147483648 Apr 29  2015 cfreds_2015_data_leakage_pc.7z.002
-rw-r--r-- 1 root root  1132827932 Apr 29  2015 cfreds_2015_data_leakage_pc.7z.003
-rw-r--r-- 1 root root 21474836480 Apr 21  2015 cfreds_2015_data_leakage_pc.dd
root@kali:~/lab#
```

Verify the unzipped DD image with MD5

```
└─$ md5sum cfreds_2015_data_leakage_pc.dd
a49d1254c873808c58e6f1bcd60b5bde  cfreds_2015_data_leakage_pc.dd
```

Step 2.

Exam partitions of the DD image using *fdisk*

```
root@kali:~/lab# fdisk -l cfreds_2015_data_leakage_pc.dd
Disk cfreds_2015_data_leakage_pc.dd: 20 GiB, 21474836480 bytes, 41943040 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf0265720

Device                          Boot  Start      End  Sectors  Size Id Type
cfreds_2015_data_leakage_pc.dd1 *      2048   206847   204800  100M  7 HPFS/NTFS/exFAT
cfreds_2015_data_leakage_pc.dd2      206848 41940991 41734144 19.9G  7 HPFS/NTFS/exFAT
root@kali:~/lab#
```

List file/directory names of the system volume

```
/bin/bash 89x27
root@kali:~/lab#
root@kali:~/lab# fls -o 206848 cfreds_2015_data_leakage_pc.dd | head
d/d 273-144-6:  Program Files (x86)
d/d 486-144-5:  Users
r/r 4-128-4:    $AttrDef
r/r 8-128-2:    $BadClus
r/r 8-128-1:    $BadClus:$Bad
r/r 6-128-4:    $Bitmap
r/r 7-128-1:    $Boot
d/d 11-144-4:   $Extend
r/r 2-128-1:    $LogFile
r/r 0-128-1:    $MFT
root@kali:~/lab#
```

If your VM doesn’t support auto-mounting

Create a folder as the mount point

```
root@kali:~/lab# mkdir /mnt/nist_dataleak_pc_dd2/
```

Mount partition 2 to the mounting point

```
root@kali:~/lab# mount /dev/loop0p2 /mnt/nist_dataleak_pc_dd2/
```

Copy HKEY_LOCAL_MACHINE (Hive) files to \lab

```
root@kali:~/lab# cp  /media/root/C8CA0C8DCA0C7A48/Windows/System32/config/DEFAULT .
root@kali:~/lab# cp  /media/root/C8CA0C8DCA0C7A48/Windows/System32/config/SAM .
root@kali:~/lab# cp  /media/root/C8CA0C8DCA0C7A48/Windows/System32/config/SECURITY .
root@kali:~/lab# cp  /media/root/C8CA0C8DCA0C7A48/Windows/System32/config/SOFTWARE .
root@kali:~/lab# cp  /media/root/C8CA0C8DCA0C7A48/Windows/System32/config/SYSTEM .
```

Verify five files

```
root@kali:~/lab# ls -l
total 21031944
-rw-r--r-- 1 root root 21474836480 Oct  3 17:50 cfreds_2015_data_leakage_pc.dd
-rwxr-xr-x 1 root root      262144 Oct  3 20:25 DEFAULT
-rwxrwxrwx 1 root root        2274 Oct  3 11:01 RegRipper30-apt-git-Install.sh
-rwxr-xr-x 1 root root      262144 Oct  3 20:25 SAM
-rwxr-xr-x 1 root root      262144 Oct  3 20:25 SECURITY
-rwxr-xr-x 1 root root    48496640 Oct  3 20:25 SOFTWARE
-rwxr-xr-x 1 root root    12582912 Oct  3 20:26 SYSTEM
```

Step 3.

Verify you have the dd image

```
/bin/bash 110x32
root@kali:~/lab#
root@kali:~/lab# ls -l cfreds_2015_data_leakage_pc.dd
-rw-r--r-- 1 root root 21474836480 Oct  3 17:50 cfreds_2015_data_leakage_pc.dd
root@kali:~/lab#
```

Compute MD5 and SHA1 of the DD image

```
root@kali:~/lab# md5sum cfreds_2015_data_leakage_pc.dd
a49d1254c873808c58e6f1bcd60b5bde  cfreds_2015_data_leakage_pc.dd
root@kali:~/lab# sha1sum cfreds_2015_data_leakage_pc.dd
afe5c9ab487bd47a8a9856b1371c2384d44fd785  cfreds_2015_data_leakage_pc.dd
root@kali:~/lab#
```

Item	Detailed Information
Filename	cfreds_2015_data_leakage_pc
MD5	A49D1254C873808C58E6F1BCD60B5BDE
SHA-1	AFE5C9AB487BD47A8A9856B1371C2384D44FD785
Imaging S/W	FTK Imager 3.4.0.1
Image Format	DD converted from VMDK (Some sectors were scrubbed[7])
Compression	Best (Smallest)
Bytes per Sector	512
Total Sectors	41,943,040
Total Size	20.00 GB (21,474,836,480 bytes)
Compressed Size	5.05 GB (5,427,795,228 bytes) ← compressed by 7zip

Show partitions of the image

```
root@kali:~/lab# fdisk -l cfreds_2015_data_leakage_pc.dd
Disk cfreds_2015_data_leakage_pc.dd: 20 GiB, 21474836480 bytes, 41943040 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf0265720

Device                          Boot  Start      End  Sectors  Size Id Type
cfreds_2015_data_leakage_pc.dd1 *      2048   206847   204800  100M  7 HPFS/NTFS/exFAT
cfreds_2015_data_leakage_pc.dd2      206848 41940991 41734144 19.9G  7 HPFS/NTFS/exFAT
root@kali:~/lab#
```

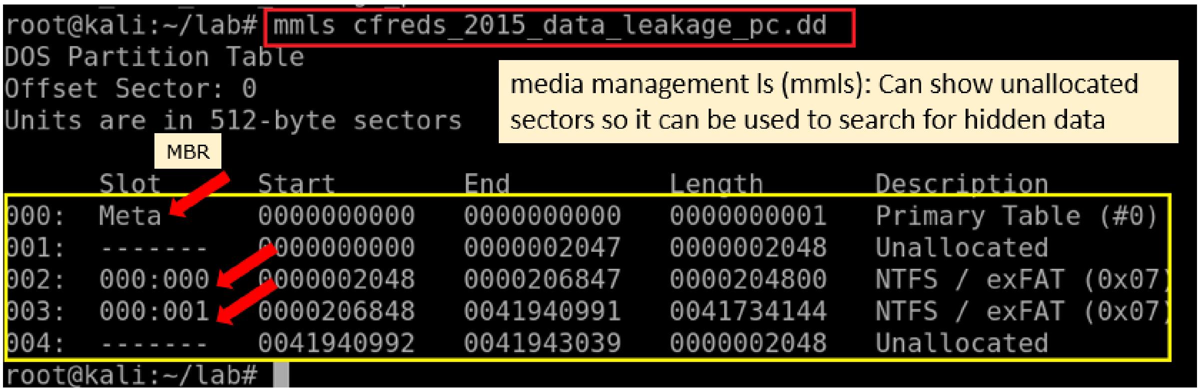
"system" volume: initial booting process and system startup

"Boot" volume: core os, load the remainder of an operating system

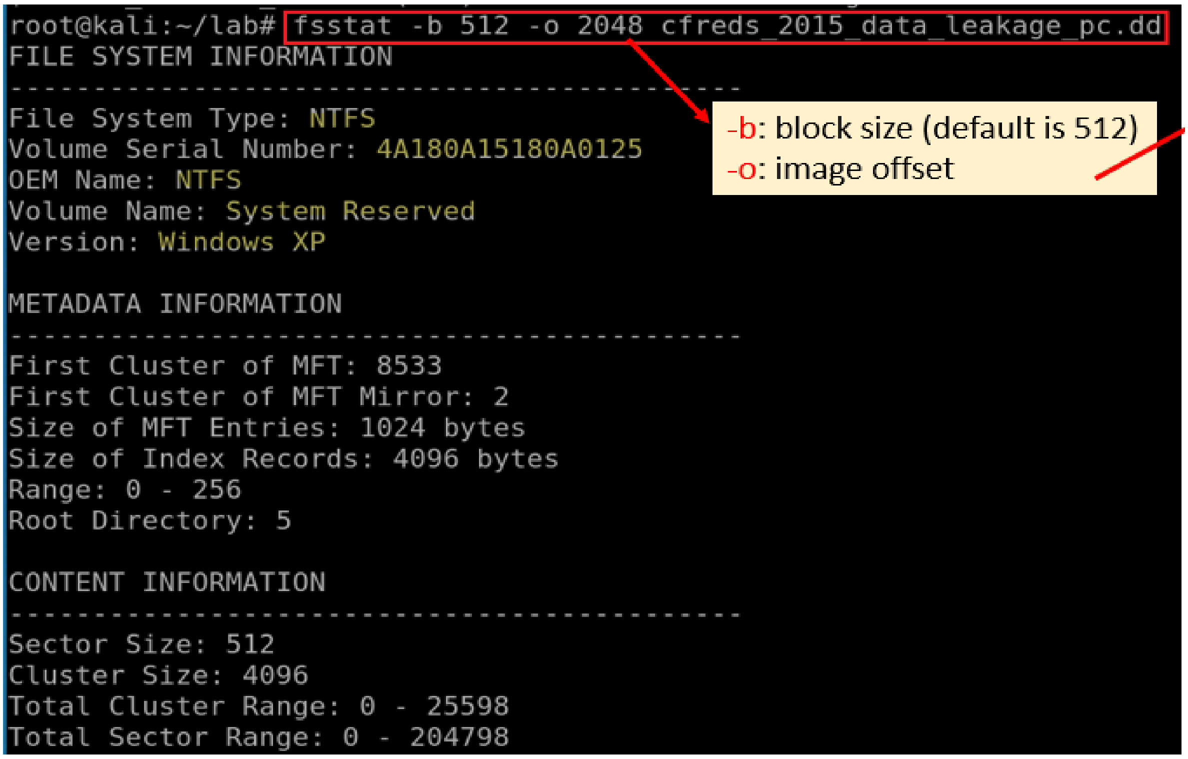
- What is Block devices? hard drives, CD-ROM drives, RAM
- What is fdisk ? Format disk

the device is bootable, not a boot partition

Show partitions and unallocated space using *mmls*



Display file system statistics and metadata information from a disk image (first partition)



List the second partition details

```
root@kali:~/lab# fsstat -b 512 -o 206848 cfreds_2015_data_leakage_pc.dd
FILE SYSTEM INFORMATION
--------------------------------------------
File System Type: NTFS
Volume Serial Number: C8CA0C8DCA0C7A48
OEM Name: NTFS
Version: Windows XP

METADATA INFORMATION
--------------------------------------------
First Cluster of MFT: 786432
First Cluster of MFT Mirror: 2
Size of MFT Entries: 1024 bytes
Size of Index Records: 4096 bytes
Range: 0 - 78080
Root Directory: 5

CONTENT INFORMATION
--------------------------------------------
Sector Size: 512
Cluster Size: 4096
Total Cluster Range: 0 - 5216766
Total Sector Range: 0 - 41734142
```

-b: block size
-o: image offset

Serial number. Remember the #. We will use it later.

List all deleted *.docx* files in the whole partition?

```
root@kali:~/lab# fls -rdF -o 206848  cfreds_2015_data_leakage_pc.dd | grep .docx
r/- * 0:    Users/informant/AppData/Roaming/Microsoft/Templates/LiveContent/15/Managed/Word
 Document Building Blocks/1033/TM02835270[[fn=Photo Sidebar (Annual Report Red and Black de
sign)]].docx
r/- * 0:    Users/informant/Desktop/~$signation_Letter_(Iaman_Informant).docx
root@kali:~/lab# 
```

Other useful parameters
- **-d** display deleted entries only
- **-D** directories only
- **-r** recursively display directories
- **-l** long format
- **-F** Display file (all non-directory) entries only
- **-u** Display undeleted entries only

Verify system information

```
root@kali:~/lab# ls -l
total 21031944
-rw-r--r-- 1 root root 21474836480 Oct  3 17:50 cfreds_2015_data_leakage_pc.dd
-rwxr-xr-x 1 root root      262144 Oct  3 20:25 DEFAULT
-rwxrwxrwx 1 root root        2274 Oct  3 11:01 RegRipper30-apt-git-Install.sh
-rwxr-xr-x 1 root root      262144 Oct  3 20:25 SAM
-rwxr-xr-x 1 root root      262144 Oct  3 20:25 SECURITY
-rwxr-xr-x 1 root root    48496640 Oct  3 20:25 SOFTWARE
-rwxr-xr-x 1 root root    12582912 Oct  3 20:26 SYSTEM
```

Verify Users' information

```
root@kali:~/lab# ls -l *.DAT
-rwxr-xr-x 1 root root  524288 Oct  3 20:58 NTUSER_Admin11.DAT
-rwxr-xr-x 1 root root  262144 Oct  3 20:59 NTUSER_Default.DAT
-rwxr-xr-x 1 root root 1048576 Oct  3 20:59 NTUSER_informant.DAT
-rwxr-xr-x 1 root root  524288 Oct  3 21:00 NTUSER_temporary.DAT
```

What is the installed OS information in detail?

```
root@kali:~/lab# rip.pl -r SOFTWARE -p winver
Launching winver v.20200525
winver v.20200525
(Software) Get Windows version & build info

ProductName                Windows 7 Ultimate
CSDVersion                 Service Pack 1
BuildLab                   7601.win7sp1_gdr.130828-1532
BuildLabEx                 7601.18247.amd64fre.win7sp1_gdr.130828-1532
RegisteredOrganization
RegisteredOwner            informant
InstallDate                2015-03-22 14:34:26Z
```

What is the computer name?

```
root@kali:~/lab# rip.pl -l | grep -i name
87. compname v.20090727 [System]
   - Gets ComputerName and Hostname values from System hive
   - Gets contents of PendingFileRenameOperations value
   - Parse hive, check key/value names for RLO character
   - Check key/value names in a hive for leading null char
root@kali:~/lab#
root@kali:~/lab# rip.pl -r SYSTEM -p compname
Launching compname v.20090727
compname v.20090727
(System) Gets ComputerName and Hostname values from System hive

ComputerName     = INFORMANT-PC
TCP/IP Hostname = informant-PC
```

How many accounts does the system have?

Search for profiles

```
root@kali:~/lab# rip.pl -l | grep -i profile
76. profilelist v.20200518 [Software]
   - Get content of ProfileList key
122. profiler v.20200525 [NTUSER.DAT, System]
   - Environment profiler information
```

```
root@kali:~/lab# rip.pl -r SOFTWARE -p profilelist
Launching profilelist v.20200518
profilelist v.20200518
(Software) Get content of ProfileList key

Microsoft\Windows NT\CurrentVersion\ProfileList

Path      : %systemroot%\system32\config\systemprofile
SID       : S-1-5-18
LastWrite : 2009-07-14 04:53:25Z

Path      : C:\Windows\ServiceProfiles\LocalService
SID       : S-1-5-19
LastWrite : 2015-03-25 11:14:18Z

Path      : C:\Windows\ServiceProfiles\NetworkService
SID       : S-1-5-20
LastWrite : 2015-03-25 11:14:18Z

Path      : C:\Users\informant
SID       : S-1-5-21-2425377081-3129163575-2985601102-1000
LastWrite : 2015-03-25 15:30:57Z

Path      : C:\Users\admin11
SID       : S-1-5-21-2425377081-3129163575-2985601102-1001
LastWrite : 2015-03-22 15:57:41Z

Path      : C:\Users\temporary
SID       : S-1-5-21-2425377081-3129163575-2985601102-1003
LastWrite : 2015-03-22 15:56:58Z
```

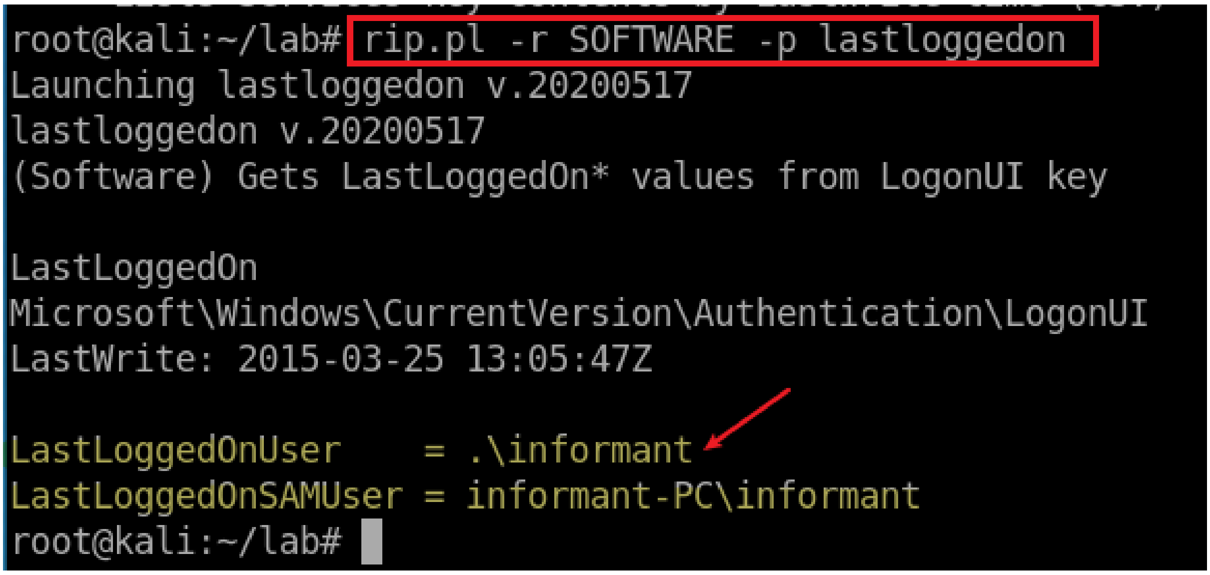
Find and search for Security Accounts Manager (SAM) information

```
root@kali:~/lab#
root@kali:~/lab# rip.pl -r SAM -P samparse | grep -E "Username|Created|Date"
Launching samparse v.20200825
Username        : Administrator [500]
Account Created : 2015-03-25 10:33:22Z
Last Login Date : 2010-11-21 03:47:20Z
Pwd Reset Date  : 2010-11-21 03:57:24Z
Pwd Fail Date   : Never
Username        : Guest [501]
Account Created : 2015-03-25 10:33:22Z
Last Login Date : Never
Pwd Reset Date  : Never
Pwd Fail Date   : Never
Username        : informant [1000]
Account Created : 2015-03-22 14:33:54Z
Last Login Date : 2015-03-25 14:45:59Z
Pwd Reset Date  : 2015-03-22 14:33:54Z
Pwd Fail Date   : 2015-03-25 14:45:43Z
Username        : admin11 [1001]
Account Created : 2015-03-22 15:51:54Z
Last Login Date : 2015-03-22 15:57:02Z
Pwd Reset Date  : 2015-03-22 15:52:10Z
Pwd Fail Date   : 2015-03-22 15:53:02Z
Username        : ITechTeam [1002]
Account Created : 2015-03-22 15:52:30Z
Last Login Date : Never
Pwd Reset Date  : 2015-03-22 15:52:45Z
Pwd Fail Date   : 2015-03-22 15:53:02Z
Username        : temporary [1003]
Account Created : 2015-03-22 15:53:01Z
Last Login Date : 2015-03-22 15:55:57Z
Pwd Reset Date  : 2015-03-22 15:53:11Z
Pwd Fail Date   : 2015-03-22 15:56:37Z
root@kali:~/lab#
```

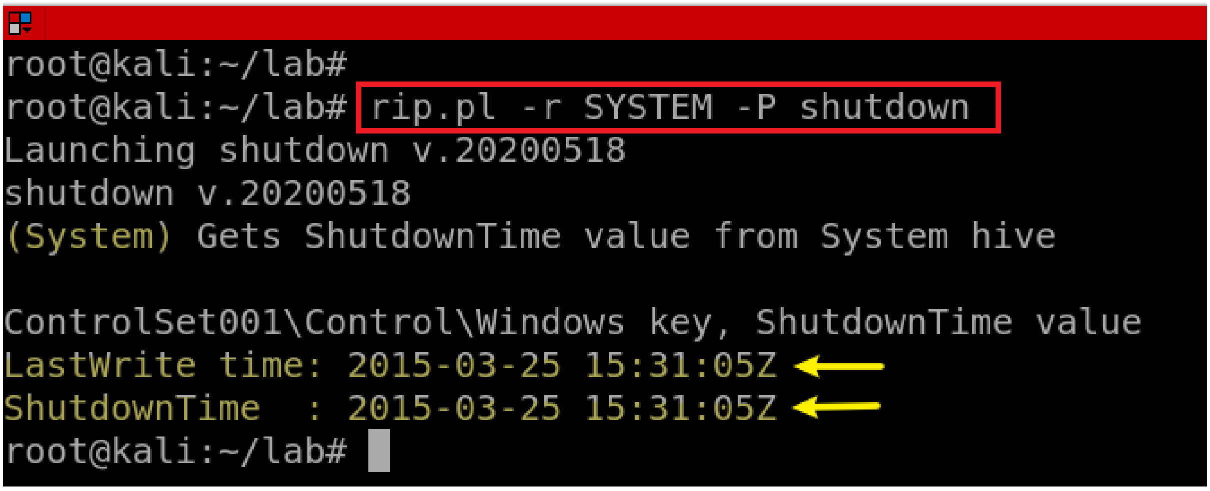
How many accounts does the system have?

When is the login time?

Who was the last user to logon into PC?



When was the last recorded shutdown date/time?



Explain the information of network interface(s) with an IP address assigned by DHCP.

```
root@kali:~/lab#
root@kali:~/lab# rip.pl -r SYSTEM -P nic2
Launching nic2 v.20200525
nic2 v.20200525
(System) Gets NIC info from System hive

Adapter: {846ee342-7039-11de-9d20-806e6f6e6963}
LastWrite Time: 2015-03-25 10:33:18Z

ControlSet001\Services\Tcpip\Parameters\Interfaces has no subkeys.
Adapter: {E2B9AEEC-B1F7-4778-A049-50D7F2DAB2DE}
LastWrite Time: 2015-03-25 15:24:51Z
  UseZeroBroadcast              0
  EnableDeadGWDetect            1
  EnableDHCP                    1
  NameServer
  Domain
  RegistrationEnabled           1
  RegisterAdapterName           0
  DhcpIPAddress                 10.11.11.129
  DhcpSubnetMask                255.255.255.0
  DhcpServer                    10.11.11.254
```

YOU MUST SUBMIT A FULL-SCREEN IMAGE FOR FULL CREDIT!

Save the document with the filename "**YOUR NAME Lab 5.pdf**", replacing "YOUR NAME" with your real name.

Email the image to the instructor as an attachment to an e-mail message. Send it to: **xxx@fe.edu.vn** with a subject line of "**Lab 5 From YOUR NAME**", replacing "YOUR NAME" with your real name.